

Allan Abraham

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Executive Summary

Software engineer with a background in Artificial Intelligence, Cloud Computing, and DevOps — developed through a specialised BCA programme at Yenepoya University in partnership with IBM. I enjoy working close to the metal: training and deploying ML models, wiring up CI/CD pipelines, and building embedded systems. I bring a problem-solver's mindset to both systems design and hands-on implementation.

Technical Skills

- **Programming Languages:** Python, Java, C/C++, JavaScript, SQL, HTML, CSS, Assembly
- **Tools & Software:** Git, Docker, GitHub Actions, Nginx, Linux Shell (Bash), Figma, STM32 HAL, VS Code
- **Databases:** PostgreSQL, MongoDB, AWS ECR/S3 Metadata Logging
- **AI/ML/Cloud:** AWS (S3, EC2, Lambda, SQS), PyTorch, OpenCV, Machine Learning, Deep Learning, NLP

Soft Skills

- Analytical Problem Solving & Systems Design
- Technical Leadership & Team Collaboration
- Systems Analysis & Troubleshooting
- Time Management & Detail-Oriented Execution
- English (Professional), Hindi (Conversational)

Professional Experience

Pothole Detection & Severity Mapping

Edge Computer Vision Project | IBM ICE & Yenepoya University, Bangalore | Jan 2026 – May 2026

- Deployed a real-time CV pipeline on a Raspberry Pi 4 — fine-tuned MobileNetV2 in PyTorch, quantized to INT8 — mounted on test vehicles to detect and log road anomalies on the fly.
- GPS frames were synced with detections via a multi-threaded logging module; spatial deduplication kept records clean, and telemetry was batch-uploaded to AWS S3.
- **Result:** Inference jumped from 4 FPS to 14 FPS (+250%), with 98% location accuracy across all geo-tagged logs.

DevOps CI/CD Automation Pipeline

Cloud & Infrastructure Project | Yenepoya University, Bangalore | Aug 2025 – Dec 2025

- Built a self-hosted CI/CD system deploying Python microservices to AWS EC2, with flake8 and pytest gates on every pull request, multi-stage Docker builds, ECR pushes, and Nginx + SSL.
- Wrote rollback scripts that triggered automatically on failed health checks, keeping production stable through every release cycle.
- **Result:** Builds finished under 4 minutes; OOM crashes on t3.micro runners eliminated; 100% production uptime maintained.

Cloud File Storage System

Full-Stack Web Project | Yenepoya University, Bangalore | May 2025 – Jul 2025

- Built a multi-user file platform with React + TailwindCSS and a FastAPI backend — JWT auth, time-limited S3 presigned URLs for uploads so credentials never touched the client.
- PostgreSQL tracked file metadata with transactional rollbacks ensuring storage and database stayed in sync at all times.
- **Result:** Zero credential exposure by design; 100% consistency between S3 and metadata.

ML Model Deployment Pipeline

Async Inference Backend | Yenepoya University, Bangalore | Nov 2024 – Apr 2025

- Designed an async inference flow: Lambda validated and queued image payloads to SQS, with multi-threaded EC2 workers pulling from the queue and running PyTorch models in parallel.
- **Result:** Peak-traffic timeouts gone; cold start dropped from 22 s (Lambda) to under 3 s (EC2 workers).

Autonomous Navigation Bot

Robotics Platform | Yenepoya University, Bangalore | Sep 2023 – May 2024

- Wrote obstacle avoidance in C++ for an Arduino Uno chassis — ultrasonic/IR sensors, non-blocking loops via `millis()`, moving-average filters for noisy readings, and an L298N H-bridge for differential steering.
- **Result:** Smooth navigation on varied surfaces; sensor noise cut by 40%, nearly eliminating false triggers.

Education

Bachelor of Computer Applications (BCA) — Artificial Intelligence, Cloud Computing & DevOps

Yenepoya University, Bangalore Campus | IBM Partnership | 2026

- 93.96% Aggregate (4557 / 4850) | CGPA 9.40 / 10 | 9.89 on 9.5-scale

Senior Secondary (12th Grade, Science – PCM)

Jaswant Modern Sr. Sec. School, Dehradun | 2021 | 92.00%

Awards & Accomplishments

- **Academic Excellence Award** — 9.89 CGPA, BCA in AI, Cloud Computing & DevOps
- **Microsoft Certified:** Power BI Data Analyst (Microsoft / Coursera)
- **AI & ML with Deep Learning and Supervised Models** (Simplilearn / Coursera)
- **Cloud Computing with AWS** (Internshala Training)
- **Fake News Detection Using Social Media Data** (TCS iON)
- **UI Design for Web Developers** (Scrimba / Coursera)
- **People & Soft Skills** (IBM / Coursera)